Buckingham Pi theorem & dimensional matrix rank

Rigorous linear-algebra proof via nullspace rank, capillary-gravity two-group dispersion, and atomic Bohr scales.

Exact statement via rank of dimensional matrix D

List all n physical quantities Q_1, ..., Q_n entering a law f(Q_1, ..., Q_n) = 0, and form the k x n exponent matrix D_ij across k base dimensions:

$$p = \dim\ker(D) = n - \operatorname{rank}(D) \Rightarrow F(\Pi_1, \Pi_2, \ \dots\ , \Pi_p) = 0$$

WHY / Rank trap proof: in static Hookean springs {F, k_spr, x}, symbols M, L, T appear (k = 3!), YET D columns are [1,1,-2], [1,0,-2], [0,1,0] with rank r = 2 -> p = 3 - 2 = 1 group Pi = F / (k_spr x)! Blind n - k = 0 would FAIL!

Two-Pi masterclass: capillary-gravity water waves

Phase speed v on deep water depends on wavenumber k, gravity g, density rho, and surface tension gamma: n = 5 variables across rank r = 3 [M, L, T] -> p = 2 qroups!

$$\Pi_1 = \frac{v}{\sqrt{g/k}}, \qquad \Pi_2 = \frac{\gamma k^2}{\rho g}\ (\mathrm{Bond}^{-1}) \Rightarrow v = \sqrt{\frac{g}{k}}\,\Phi\!\left(\frac{\gamma k^2}{\rho g}\right)$$

Exact hydrodynamics supplies Phi(u) = sqrt(1 + u), giving v = sqrt(g/k + gamma k / rho) with minimum speed v_min = (4 g gamma / rho)^(1/4) = 23.1 cm/s on water!

WHY / Without fluid PDEs, Pi theorem ALONE proves minimum speed scales as (g gamma / rho)^(1/4) at wavelength ~ sqrt(gamma / rho g)!

Hydrogen Bohr radius & Rydberg energy (OL8, P34)

Non-relativistic quantum electrostatics has 4 independent parameters {m_e, e, eps0, hbar} spanning all 4 dimensions [M, L, T, A] (rank r = 4):

$$a_0 = \frac{4\pi\varepsilon_0\hbar^2}{m_e e^2} = 0.0529\,\mathrm{nm}, \qquad |E_1| = \frac{m_e e^4}{32\pi^2\varepsilon_0^2\hbar^2} = 13.6\,\mathrm{eV}$$

WHY / Notice: adding speed of light c introduces n = 5th variable at rank 4 -> p = 1 dimensionless group alpha = e^2/(4 pi eps0 hbar c) ~ 1/137 (fine-structure splitting ~ alpha^2 |E_1|)!

Trinity nuclear test & prefactor caution (OL9, P28)

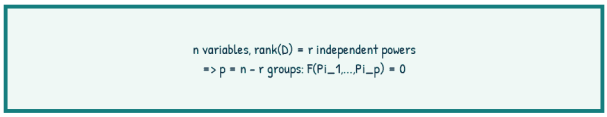
For Trinity 1945 (E = 84 TJ, t = 0.025 s, R = 100 m), naive inversion rho ~ E t^2 / R^5 (setting k = 1 in R = k (E t^2 / rho)^(1/5)) gives:

$$\frac{Et^2}{R^5} = \frac{(84\times10^{12})(0.025)^2}{100^5} = 5.25\,\mathrm{kg/m^3}\ \ (\mathrm{air:}\ \ 1.25)$$

WHY / Why off by 4.2x? Because k enters raised to the FIFTH POWER: G. I. Taylor solved the adiabatic shock ODE for air (gamma = 1.4) to get k = 1.033 (1.033^5 = 1.18; remaining factor is yield calibration)!

WATCH / Whenever solving backwards for a quantity inside a 1/5 power, a tiny 20% uncertainty in k or R explodes by factor (1.2)^5 = 2.5x!

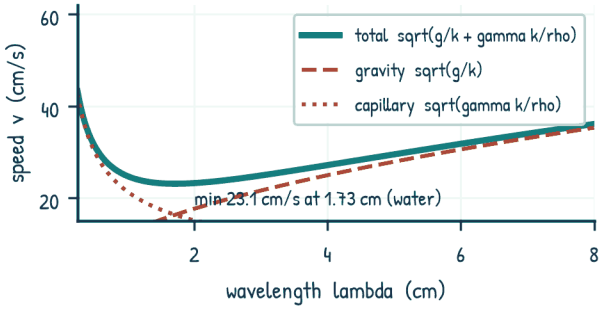
Buckingham Pi theorem via rank of the exponent matrix D (nullspace dimension n - r)



WATCH: r = rank(D), NOT blind symbol count (e.g. M T^-2 locked)!

Bohr (E, m_e, e, eps0, hbar): n=5, r=4 => p=1 => E ~ m_e e^4/(eps0^2 hbar^2)

Two dimensionless groups (n=5, r=3 -> p=2): crossover at capillary length 2 pi sqrt(gamma/rho g)



REMEMBER / Number of independent dimensionless groups is ALWAYS p = n - rank(D), NEVER blind n - k when two base dimensions only enter locked together!

Dynamical similarity, scale models & allometry

Matching dimensionless groups in wind tunnels & towing tanks, conflict of simultaneous Re + Fr matching, and Galileo scaling.

Four master dimensionless numbers of continuum physics

Divide competing force terms in Navier-Stokes / elasticity to extract universal regime numbers without solving the PDE:

$$\mathrm{Re} = \frac{\rho v L}{\eta}\ \left(\frac{\text{inertial}}{\text{viscous}}\right), \qquad \mathrm{Fr} = \frac{v}{\sqrt{gL}}\ \left(\sqrt{\frac{\text{inertial}}{\text{gravity}}}\right)$$

$$\mathrm{Ma} = \frac{v}{c_s}\ \left(\frac{\text{flow}}{\text{sound}}\right), \qquad \mathrm{We} = \frac{\rho v^2 L}{\gamma}\ \left(\frac{\text{inertial}}{\text{capillary}}\right)$$

WHY / Sphere drag F = (1/2) C_D(Re) rho pi r^2 v^2 unifies BOTH regimes: for Re << 1, C_D = 24/Re recovers linear Stokes drag 6 pi eta r v; for Re ~ 10^4, C_D ~ 0.47 gives quadratic drag!

Wind-tunnel & ship-hull model scaling dilemma

Test a 1:25 scale aircraft wing (L_m / L_p = 1/25) or ship hull in the laboratory by keeping Pi_model = Pi_prototype:

$$\text{Re match}: \ v_m = 25 v_p \ \text{ or pressurise } \ \rho_m = 25\rho_p$$

$$\text{Ship waves (Froude match)}: \ \frac{v_m}{\sqrt{gL_m}} = \frac{v_p}{\sqrt{gL_p}} \Rightarrow v_m = \frac{v_p}{5}$$

WHY / Classic IPhO trap: in the SAME fluid at fixed g, Re requires v_m ~ L^-1 while Fr requires v_m ~ L^+1/2 -> BOTH can match ONLY in a fluid of kinematic viscosity nu_m = nu_p (L_m/L_p)^(3/2)!

Mechanical & gravitational self-similarity

Scale every linear dimension of a structure or planet uniformly by factor lambda (L -> lambda L) at fixed material density rho and strength sigma_max:

$$M \propto \lambda^3, \qquad F_{\text{area}} \propto \lambda^2, \qquad \sigma_{\text{own weight}} = \frac{Mg}{A_{\text{bone}}} \propto \frac{\lambda^3}{\lambda^2} = \lambda^1$$

WHY / Galileo square-cube law: doubling an animal height without thickening proportions doubles bone stress! Constant safety factor requires bone diameter d ~ L^(3/2)!

Self-gravitating homogeneous sphere: surface g = (4/3) pi G rho R ~ R^1, free-fall / orbital period T_orb = sqrt(3 pi / G rho) ~ R^0 (INDEPENDENT of planet size R)!

Jumping animals, cooling rates & heat loss scaling

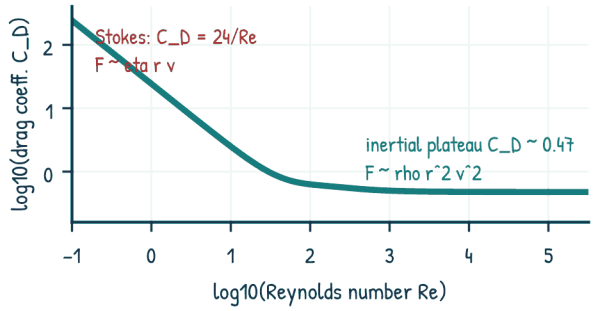
Compare maximal vertical jump height h of geometrically similar animals (muscle work W_muscle = sigma_muscle * A_muscle * s_stroke ~ L^2 * L = L^3!):

$$mgh \sim W_{\text{muscle}} \Rightarrow (\lambda^3) gh \sim \lambda^3 \Rightarrow h \propto \lambda^0$$

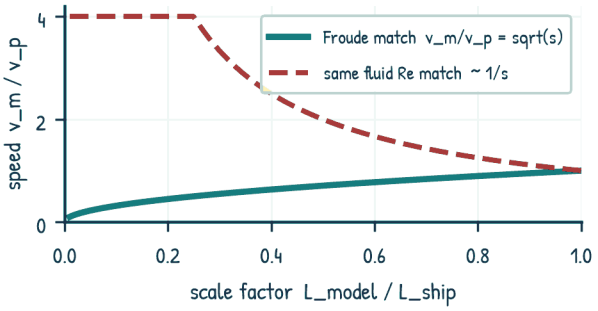
WHY / Heat-loss timescale of a warm body of heat capacity M c ~ L^3 losing heat across skin area A ~ L^2 scales as tau_cool ~ L^1: small shrews must eat continuously!

WATCH / Always state which material parameters stay constant during scaling (e.g. constant density rho vs constant total mass M changes every exponent)!

Sphere drag across six decades of Re = rho v d / eta bridges Stokes to quadratic drag



Why ship tanks cannot match Froude AND Reynolds simultaneously in water



REMEMBER / A scale model reproduces full non-linear physics faithfully IF AND ONLY IF every relevant dimensionless Pi group has the exact same numerical value!

Least-squares regression, normal equations & errors

Exact derivation of OLS slope/intercept, standard errors sigma_b and sigma_a, x-vs-y asymmetry, and weighted fits.

Normal equations & centroid theorem (OL1, P18)

For n pairs (x_i, y_i) with exact x_i and independent equal-variance vertical errors sigma_y, minimise S(a, b) = sum_i (y_i - a - b x_i)^2:

$$b = \frac{S_{xy}}{S_{xx}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \qquad a = \bar{y} - b\bar{x}$$

WHY / Setting dS/da = 0 immediately forces y_bar = a + b x_bar: EVERY unconstrained best-fit line passes strictly through centroid (x_bar, y_bar), NOT (0, 0)!

WATCH / Regressing x on y gives slope b_xy = S_xy / S_yy != 1 / b_yx: their product b_yx * b_xy = r^2 <= 1 equals squared Pearson correlation!

Standard errors of slope sigma_b & intercept sigma_a (OL2)

Notice b = sum_i c_i y_i is linear in y_i with weights c_i = (x_i - x_bar)/S_xx satisfying sum c_i^2 = 1/S_xx. Propagating variance gives:

$$\sigma_y = \sqrt{\frac{\sum r_i^2}{n-2}}, \qquad \sigma_b = \frac{\sigma_y}{\sqrt{S_{xx}}}, \qquad \sigma_a = \sigma_y \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}$$

WHY / Look at n - 2 in residual scatter sigma_y: fitting TWO parameters (a, b) consumes 2 degrees of freedom (with just n = 2 points, line hits both exactly so residuals = 0)!

Worked full Olympiad exam drill (Paper P30)

Data {(1, 2.1), (2, 3.9), (3, 6.2), (4, 7.8)} with known sigma_y = 0.2: here x_bar = 2.5, y_bar = 5.0, dx = {-1.5, -0.5, +0.5, +1.5}:

$$S_{xx} = 5.0, \;\; S_{xy} = 9.70 \Rightarrow b = 1.94, \qquad a = 0.15$$

Uncertainties: sigma_b = 0.2 / sqrt(5) = 0.089, sigma_a = 0.2 * sqrt(0.25 + 6.25/5.0) = 0.2 * sqrt(1.50) = 0.245 -> y = (0.15 +- 0.25) + (1.94 +- 0.09) x!

WHY / Design rule: placing half your trials near x_min and half near x_max maximizes S_xx (after verifying linearity with interior points)!

Through-origin slope & unequal error bars (WLS)

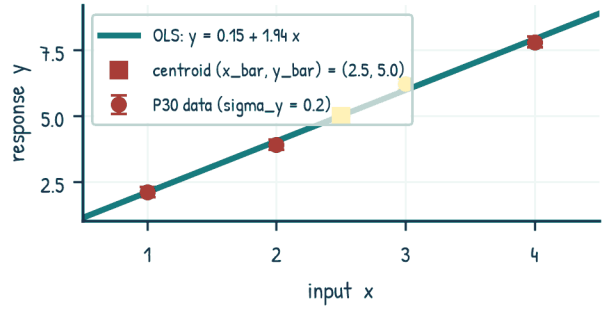
1. If physics GUARANTEES zero intercept (y = b x with zero offset verified), minimising sum (y_i - b x_i)^2 gives:

$$b_{\text{origin}} = \frac{\sum x_i y_i}{\sum x_i^2}, \qquad \sigma_{b,\,\text{origin}} = \frac{\sigma_y}{\sqrt{\sum x_i^2}}$$

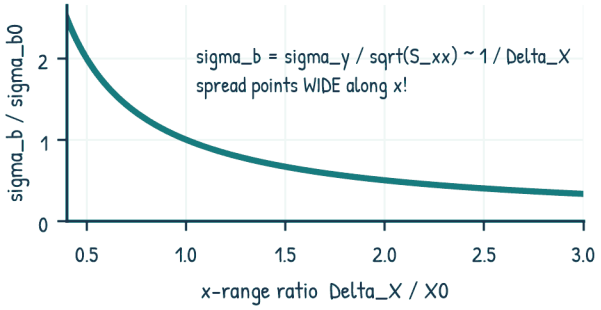
2. When error bars sigma_i vary from point to point (e.g. after taking ln y!), weight every point by inverse variance w_i = 1 / sigma_i^2:

$$w_i = \sigma_i^{-2} \Rightarrow b = \frac{\sum w_i (x_i - \bar{x}_w)(y_i - \bar{y}_w)}{\sum w_i (x_i - \bar{x}_w)^2}$$

Paper P30 fit: OLS line passes strictly through centroid (2.5, 5.0) with b = 1.94 +- 0.09



Design lever for straight-line slopes: doubling x-span halves slope uncertainty sigma_b



REMEMBER / OLS line ALWAYS passes through centroid (x_bar, y_bar) with b = S_xy / S_xx and sigma_b = sigma_y / sqrt(S_xx) — spread x_i wide!

Chi-square goodness of fit, weighted mean & covariance

Testing models quantitatively with chi^2/(n - p), optimal inverse-variance data fusion, and full correlated error propagation.

Chi-square test statistic & degrees of freedom (OL3)

For n independent Gaussian measurements y_i +- sigma_i fitted by a theoretical curve f(x_i; theta_1, ..., theta_p) having p adjustable parameters:

$$\chi^2 = \sum_{i=1}^{n} \left(\frac{y_i - f(x_i)}{\sigma_i} \right)^2, \qquad \nu = n - p, \qquad \chi^2_\nu \equiv \frac{\chi^2}{\nu}$$

WHY / Each standardised residual z_i = r_i / sigma_i ~ N(0, 1) contributes E[z_i^2] = 1, minus p fitted constraints: E[chi^2] = nu and SD(chi^2) = sqrt(2 nu)!

Check P30(c): residuals {+0.01, -0.13, +0.23, -0.11} with sigma = 0.2 give chi^2 = 2.05 for nu = 4 - 2 = 2 (chi^2_nu = 1.03 ~ 1 -> textbook fit!).

Three diagnostic outcomes of reduced chi^2_nu

1. chi^2_nu ~ 1 (within 1 +- sqrt(2/nu)): model AND stated uncertainties sigma_i are mutually consistent.

2. chi^2_nu >> 1 (e.g. chi^2 = 45 for nu = 8): EITHER model is missing real physics (curvature / systematic drift) OR error bars sigma_i were underestimated!

WATCH / 3. chi^2_nu << 1 (e.g. chi^2 = 0.4 for nu = 10): NOT "miraculously accurate data" — your error bars sigma_i are way too conservative (or data was overfitted/fudged)!

Optimal combination of independent labs (weighted mean)

Two or more independent experiments measure the SAME physical constant with different precisions x_1 +- sigma_1, ..., x_k +- sigma_k. Minimising chi^2(mu) = sum (x_i - mu)^2 / sigma_i^2 gives:

$$w_i = \frac{1}{\sigma_i^2}, \qquad \bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}, \qquad \sigma_{\bar{x}_w} = \frac{1}{\sqrt{\sum w_i}} < \min_i \sigma_i$$

WHY / Combined precision ALWAYS beats every individual experiment! If sigma_1 = 0.01 and sigma_2 = 0.10, Run 1 carries weight 100x larger than Run 2 — NEVER take unweighted mean (x1+x2)/2!

Correlated errors & covariance matrix formula

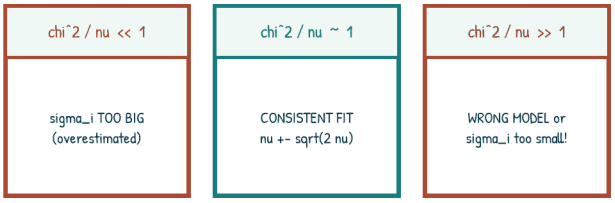
Whenever two fitted quantities (such as OLS intercept a and slope b, with Cov(a, b) = -x_bar sigma_y^2 / S_xx < 0!) derive from the SAME dataset, use covariance C_xy = <dx dy>:

$$\sigma_f^2 = \left(\frac{\partial f}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \sigma_y^2 + 2\left(\frac{\partial f}{\partial x}\right)\left(\frac{\partial f}{\partial y}\right) \mathrm{Cov}(x, y)$$

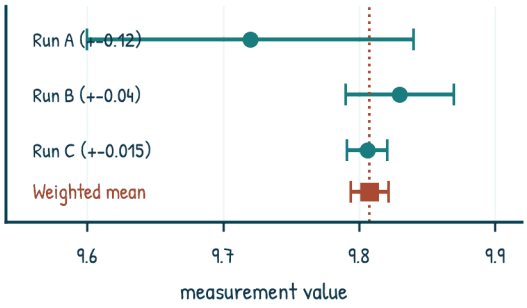
WHY / Look how nicely this explains extrapolating y_hat(x) = a + b x: plug Cov(a, b) in and cross-term cancels x_bar to give exact confidence fan:

$$\sigma_{\hat{y}(x)} = \sigma_y \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}} \quad (\text{tightest right at centroid } x = \bar{x}!)$$

Diagnostic meaning of reduced chi-square chi^2_nu = chi^2 / (n - p)



Inverse-variance weighting w_i = 1/sigma_i^2 pulls combined estimate to sharpest run



REMEMBER / A trustworthy fit of p parameters to n data points satisfies chi^2 ~ (n - p) +- sqrt(2(n - p)); combining measurements weights strictly by 1/sigma_i^2!

Poisson counting statistics, background & dead time

Exact sqrt(N) shot noise, rigorous background-subtraction formula (fixing P35c), optimal time splitting, and Geiger dead time.

Poisson event counts & the universal sqrt(N) law (OL6)

Independent random point events at mean rate R over time t follow Poisson law P(k) = mu^k exp(-mu) / k! with mean mu = R t and exact standard deviation:

$$\sigma_N = \sqrt{N}, \qquad \frac{\sigma_N}{N} = \frac{1}{\sqrt{N}}, \qquad R = \frac{N}{t} \pm \frac{\sqrt{N}}{t} = R \pm \sqrt{\frac{R}{t}}$$

Worked OL6: N = 10,000 counts in 600 s gives 1% precision (16.67 +- 0.17 s^-1). Reaching 0.1% demands N = 10^6 counts -> 100x longer run (16.7 hours)!

WATCH / CRITICAL TRAP: apply sqrt(.) ONLY to raw integer COUNTS N, NEVER to rate R! If 100 counts arrive in 10 s, R = 10 +- 1 s^-1 (NOT 10 +- sqrt(10) = 10 +- 3.16)!

Exact background subtraction (audit of Paper P35c!)

Count source+background for t_on (observing N_tot = R_tot t_on) and pure background separately for t_off (observing N_b = R_b t_off):

$$R_{\text{net}} = \frac{N_{\text{tot}}}{t_{\text{on}}} - \frac{N_b}{t_{\text{off}}}, \qquad \sigma(R_{\text{net}}) = \sqrt{\frac{R_{\text{tot}}}{t_{\text{on}}} + \frac{R_b}{t_{\text{off}}}} = \sqrt{\frac{R_s + R_b}{t_{\text{on}}} + \frac{R_b}{t_{\text{off}}}}$$

WHY / Audit note on P35(c): with R_tot = 250/s (R_s = 200/s) and R_b = 50/s, if R_b is known with zero error then sigma(R_net) = sqrt(2500)/10 = 5.0/s; ONLY when R_b is ALSO measured for 10 s (N_b = 500) does variance become (2500 + 500)/10^2 -> 5.48/s!

Optimal beam-time split for fixed total lab time T

Given fixed total detector time T = t_on + t_off, minimise variance V = R_tot / t_on + R_b / (T - t_on) with respect to t_on:

$$-\frac{R_{\text{tot}}}{t_{\text{on}}^2} + \frac{R_b}{t_{\text{off}}^2} = 0 \Rightarrow \frac{t_{\text{on}}}{t_{\text{off}}} = \sqrt{\frac{R_{\text{tot}}}{R_b}}$$

WHY / Clean physical limits: if signal >> background (R_tot >> R_b), spend almost all time on source; if weak signal sits atop huge background (R_tot ~ R_b), split 50% / 50%!

Detector dead time tau (non-paralysable model)

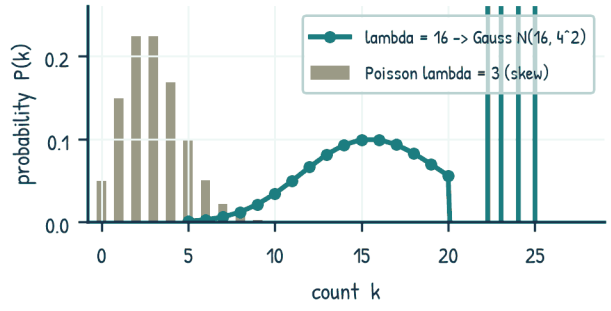
After each registered click, a Geiger/scintillation counter stays blind for dead time tau. In 1 second it registers m clicks and is dead for total fraction m * tau:

$$m = R_{\text{true}}(1 - m\tau) \Rightarrow R_{\text{true}} = \frac{m}{1 - m\tau}, \qquad m = \frac{R_{\text{true}}}{1 + R_{\text{true}}\tau}$$

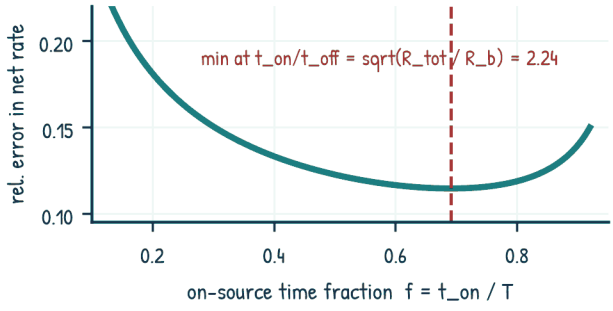
WHY / At intense rates R_true >> 1/tau, observed count rate saturates at ceiling 1/tau AND sub-Poisson anti-bunching breaks naive sqrt(N) statistics!

Two-source lab calibration: measure m1, m2 separately and m12 together -> m1 + m2 > m12 reveals dead time tau directly without knowing R_true!

Poisson counts converge to Gaussian mean N and standard deviation sqrt(N) for N >> 10



Optimal beam-time allocation between gross rate R_tot = 250/s and background R_b = 50/s



REMEMBER / Subtracting background REQUIRES ADDING variances: sigma(R_net) = sqrt(R_tot / t_on + R_b / t_off); NEVER substitute R_net inside the square root!

Olympiad experiment design: 0.1% g-budget & natural units

End-to-end error budget engineering (OL5 / P32), systematic corrections on real pendulums, and symmetry scaling laws.

Engineering a 0.10% measurement of g (OL5, P32)

Available gear: ruler (dL = 1 mm), stopwatch (dt = 0.01 s), string up to 2 m, steel bob. Target: dg / g < 0.10% from g = 4 pi^2 L / T^2:

$$\frac{\Delta g}{g} = \frac{\Delta L}{L} + 2\frac{\Delta t_{\text{sw}}}{NT} + 2\left(\frac{\theta_0^2}{16}\right)$$

Design choices: 1. Max out length L = 2.000 m so dL/L = 1/2000 = 0.050%. 2. Now T = 2.84 s; time N = 50 continuous swings (142 s) so 2 dt/(N T) = 0.014%!

3. Restrict launch angle to theta_0 = 2 deg = 0.0349 rad so 2(theta_0^2/16) = 0.015%! Total budget = 0.050 + 0.014 + 0.015 = 0.079% < 0.10%!

Three hidden physical systematics in real pendulums

To trust 0.1% accuracy (not merely precision!), account for three real-world physics corrections beyond point-mass vacuum theory:

$$I_{\text{pivot}} = mL^2 + \tfrac{2}{5}mr_{\text{bob}}^2 \Rightarrow L_{\text{eff}} = L_{\text{cm}}\left(1 + \frac{2r_{\text{bob}}^2}{5L_{\text{cm}}^2}\right)$$

$$g_{\text{eff}} = g\left(1 - \frac{\rho_{\text{air}}}{\rho_{\text{bob}}}\right) \quad (0.015\%\ \text{steel},\ 0.05\%\ \text{glass})$$

WHY / Plus time zero-crossings AT EQUILIBRIUM (fastest bob speed -> sharpest visual trigger!), NEVER at extreme turning points where v = 0!

Relativistic corrections via symmetry + dimensions (OL10, P36)

How must momentum p = m v f(v/c) or rod length L = L0 g(v/c) behave at small speeds beta = v/c BEFORE invoking full Lorentz algebra?

$$\mathbf{p} = m\mathbf{v}\, f\left(\frac{\mathbf{v}\cdot\mathbf{v}}{c^2}\right) = m\mathbf{v}\left(1 + \alpha_1\frac{v^2}{c^2} + O(\beta^4)\right)$$

WHY / Spatial isotropy + smoothness at v = 0 forbids odd linear terms +-v/c in scalar factor f! Special relativity fixes alpha_1 = +1/2 for momentum (1% shift at v = 0.141 c, P36) and -1/2 for Lorentz contraction!

Natural units c = hbar = k_B = 1 (theoretical Olympiad tool)

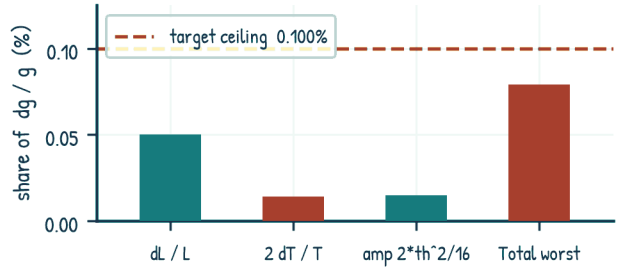
In particle/astrophysics Olympiad problems using natural units (hbar = c = k_B = 1), EVERY mechanical/thermal quantity collapses onto powers of ENERGY [E]:

$$[M] = [T_{\text{temp}}] = [p] = [E], \quad [L] = [t] = [E^{-1}], \quad [A] = [E^{-2}]$$

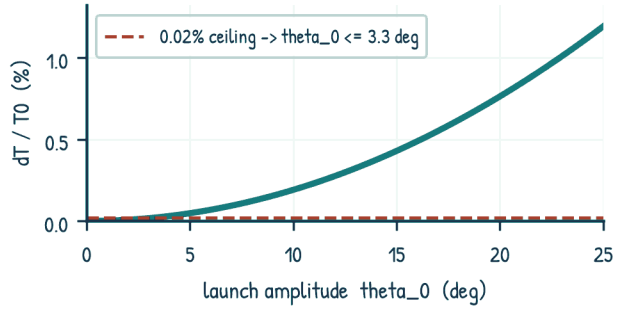
WHY / To restore hbar and c at the end of any derivation, uniquely match [M, L, T]: e.g. energy density [u] = [E^4] restores to u ~ (k_B T)^4 / (hbar c)^3 (Stefan-Boltzmann T^4 scaling in ONE second)!

Handy anchor: hbar c = 197.3 MeV fm = 197.3 eV nm!

OL5 experimental budget (L = 2.000 m, N = 50 swings, theta_0 = 2 deg): 0.079% < 0.1%!



Anharmonic pendulum period drift +theta_0^2 / 16 (theta_0 MUST be in radians!)



REMEMBER / In experiment design, list EVERY random AND systematic term in dg/g BEFORE touching equipment, and engineer every term well below ceiling!